

GORIN, Russian Federation

WMO: 314890

Lat: 51.1931N

Lon: 136.6500E

Elev: 301

StdP: 14.54

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-30.6	-40.3	0.5	-34.5	-36.0	0.7	-30.1	15.4	10.8	12.3	9.9	0.4	210	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	86.1	71.3	82.7	69.2	79.6	67.7	73.9	82.8	71.7	80.3	69.7	77.3	3.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.9	115.3	79.2	68.7	106.6	76.4	66.8	99.9	74.3	37.8	82.8	35.8	80.4	34.0	77.7	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	10.0	8.8	-39.5	91.7	3.7	3.3	-42.1	94.1	-44.3	96.0	-46.4	97.9	-49.0	100.3
			-39.7	76.8	3.5	2.0	-42.3	78.2	-44.3	79.4	-46.3	80.6	-48.9	82.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.3	-11.4	-2.7	15.0	35.1	50.4	60.7	67.9	65.1	53.5	36.7	11.2	-8.0		
	DBStd	29.46	10.91	9.72	9.72	8.55	8.05	7.17	6.37	5.28	6.82	7.48	12.80	11.19		
	HDD50	8442	1904	1476	1085	454	97	5	0	1	42	415	1165	1798		
	HDD65	12539	2369	1896	1550	896	454	164	43	63	349	877	1615	2263		
	CDD50	1620	0	0	0	8	111	328	555	469	146	3	0	0		
	CDD65	241	0	0	0	0	2	37	133	66	3	0	0	0		
	CDH74	2323	0	0	0	5	94	515	1172	497	40	0	0	0		
	CDH80	626	0	0	0	0	14	123	401	87	1	0	0	0		
(14) Wind	WSAvg	3.1	2.3	2.7	3.8	4.2	4.0	2.9	2.6	2.5	2.7	3.4	3.2	2.6		
(15) Precipitation	PrecAvg	25.2	0.7	0.6	0.8	1.7	2.5	2.6	4.0	4.3	2.7	2.1	1.2	1.1		
	PrecMax	33.0	2.7	2.5	1.9	8.0	5.4	5.2	7.2	10.6	6.4	5.7	4.0	2.4		
	PrecMin	14.2	0.1	0.1	0.1	0.2	0.2	0.6	0.0	0.7	0.6	0.4	0.1	0.0		
	PrecStd	5.0	0.6	0.5	0.5	1.7	1.4	1.0	1.9	2.6	1.4	1.5	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	26.9	45.0	71.0	81.8	87.9	92.7	85.9	78.6	65.3	44.0	25.1		
		MCWB	23.0	22.7	35.5	52.5	61.7	71.1	75.5	72.1	64.9	52.0	36.2	23.6		
	2%	DB	17.2	21.1	39.0	62.4	76.1	83.6	88.1	82.3	73.9	57.5	37.4	19.9		
		MCWB	15.5	18.5	31.5	47.4	57.0	67.7	73.0	70.0	62.5	46.5	32.6	18.5		
	5%	DB	11.2	17.2	34.9	56.3	72.1	80.1	84.4	79.5	70.0	52.8	32.8	14.8		
		MCWB	9.8	15.0	29.2	43.5	54.9	65.7	71.7	69.0	60.1	43.6	29.2	13.4		
	10%	DB	6.1	13.8	30.7	50.6	67.4	76.4	80.8	76.4	65.9	48.9	29.5	9.5		
		MCWB	5.0	11.9	26.2	40.3	52.9	63.8	70.0	67.3	57.6	41.6	26.8	8.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	22.9	35.6	54.4	63.2	72.6	79.2	75.3	67.2	52.8	36.6	23.9		
		MCDB	24.7	26.3	43.0	69.4	79.1	84.2	86.6	81.8	76.0	62.8	42.2	24.9		
	2%	WB	15.6	18.5	32.2	48.3	59.2	69.8	75.2	72.8	64.2	48.6	33.0	18.4		
		MCDB	17.0	21.1	38.3	61.6	71.8	81.1	85.3	79.9	72.0	54.6	36.7	19.8		
	5%	WB	9.9	15.0	29.4	44.1	56.5	67.7	72.9	70.6	61.5	45.0	29.9	13.4		
		MCDB	11.1	17.0	34.4	55.2	69.2	77.5	82.3	77.5	67.8	51.2	32.6	14.7		
	10%	WB	5.0	12.0	26.5	40.5	53.9	65.1	70.9	68.4	59.1	42.2	26.7	8.3		
		MCDB	6.1	13.9	30.6	50.0	64.3	73.5	79.0	74.4	64.3	47.7	29.4	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	23.4	22.1	19.2	21.7	21.1	18.2	17.7	20.0	18.1	17.6	16.7		
		MCDBR	14.9	22.7	25.0	30.2	33.0	30.0	25.7	24.7	27.1	25.8	18.6	14.5		
		MCWBR	14.1	19.9	18.3	16.8	16.2	15.3	12.3	13.5	15.8	15.5	14.1	13.7		
		MCDBR	14.6	21.9	22.6	28.9	29.1	25.1	22.9	21.2	24.3	23.9	16.3	13.8		
	5% WB	MCWBR	13.9	19.1	16.6	16.6	15.2	13.7	12.3	13.1	15.4	15.6	12.7	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.303	0.375	0.486	0.565	0.500	0.497	0.423	0.381	0.392	0.328	0.271		
	taud		2.112	2.045	1.886	1.800	1.759	2.023	2.082	2.308	2.350	2.122	2.080	2.094		
	Ebn at Noon		232	254	252	237	224	243	241	256	254	217	200	207		
	Edh at Noon		24	35	50	62	67	52	49	37	31	32	25	21		
(44) All-Sky Solar Radiation	RadAvg		481	868	1380	1644	1612	1780	1626	1408	1105	698	437	365		
	RadStd		19	29	59	121	144	176	164	128	87	54	38	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page